SPNet: Siamese-Prototype Network for Few-Shot Remote Sensing Image Scene Classification

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***Abstract*— Few-shot image classification has attracted extensive attention, which aims to recognize unseen classes given only a few labeled samples. Due to the large intraclass variances and interclass similarity of remote sensing scenes, the task under such circumstance is much more challenging than general few-shot image classification. Most existing prototype-based few-shot algorithms usually calculate prototypes directly from support samples and ignore the validity of prototypes, which results in a decline in the accuracy of subsequent inferences based on prototypes. To tackle this problem, we propose a Siamese-prototype network (SPNet) with prototype self-calibration (SC) and intercalibration (IC). First, to acquire more accurate prototypes, we utilize the supervision information from support labels to calibrate the prototypes generated from support features. This process is called SC. Second, we propose to consider the confidence scores of the query samples as another type of prototypes, which are then used to predict the support samples in the same way. Thus, the information interaction between support and query samples is implicitly a further calibration for prototypes (so-called IC). Our model is optimized with three losses, of which two additional losses help the model to learn more representative prototypes and make more accurate predictions. With no additional parameters to be learned, our model is very lightweight and convenient to employ. The experiments on three public remote sensing image datasets demonstrate competitive performance compared with other advanced few-shot image classification approaches. The source code is available at https://github.com/zoraup/SPNet.**

***Index Terms*— Few-shot learning, remote sensing image scene classification, Siamese-prototype network (SPNet).**

I. Introduction

REMOTE sensing image scene classification has been developing these years rapidly as it has a wide range of applications in the real world, such as natural hazards detection [1], [2], land-use classification [3]–[5], and urban planning [6], [7].

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Remote sensing image scene classification aims to assign a scene category to each remote sensing image, such as "airport," "forest," and "industrial area." In recent years, deep learning has been a powerful means to accomplish many tasks in image and video processing [8]–[12]. In 2015, Penatti *et al.* [13] first proposed to use convolutional neural networks (CNNs) to deal with remote sensing image scene classification. They verified that the high-level features extracted by CNNs were more effective than the traditional low-level features, whereafter a great deal of CNN-based scene classification methods appeared [14]–[18].

However, there are still two essential problems in the scene classification of remote sensing images. First, although traditional remote sensing image scene classification methods can predict the test samples of the scene categories seen during the training phase with quite high accuracies, it may be stranded when confronted with the samples whose categories have never been seen during the training phase. Second, it is known that the training process in traditional deep learning methods requires a massive amount of labeled data, which is hard to obtain in a novel and unknown environment. Under the circumstances, few-shot learning was applied to break these predicaments.

As the name implies, few-shot learning aims to learn new concepts from a few annotated examples [19], [20]. It has been actively explored in recent years. On the whole, few-shot learning methods can be divided into three categories: model-based, metric-based, and optimization-based. Model-based methods aim to update parameters quickly on a small number of samples through the design of the model structure and directly establish the mapping function of input and predicted value [21]–[23]. Metric-based methods utilize the idea of the nearest neighbor to complete the classification by measuring the distance between the samples in the query set and support set [24]–[26]. Optimization-based methods consider that the common gradient descent methods are difficult to fit the few-shot scenario, so the task is carried out by adjusting the optimization method [27]–[30].

Among these methods, metric-based methods are mostly studied in the past few years in both natural image classification and remote sensing image scene classification [24]–[26], [31]–[38]. Recently, Li *et al.* [33] introduced meta-learning to remote sensing image scene classification. Following closely behind, Li *et al.* [36] adopted the attention technique to delve into the interchannel and interspatial relationships to

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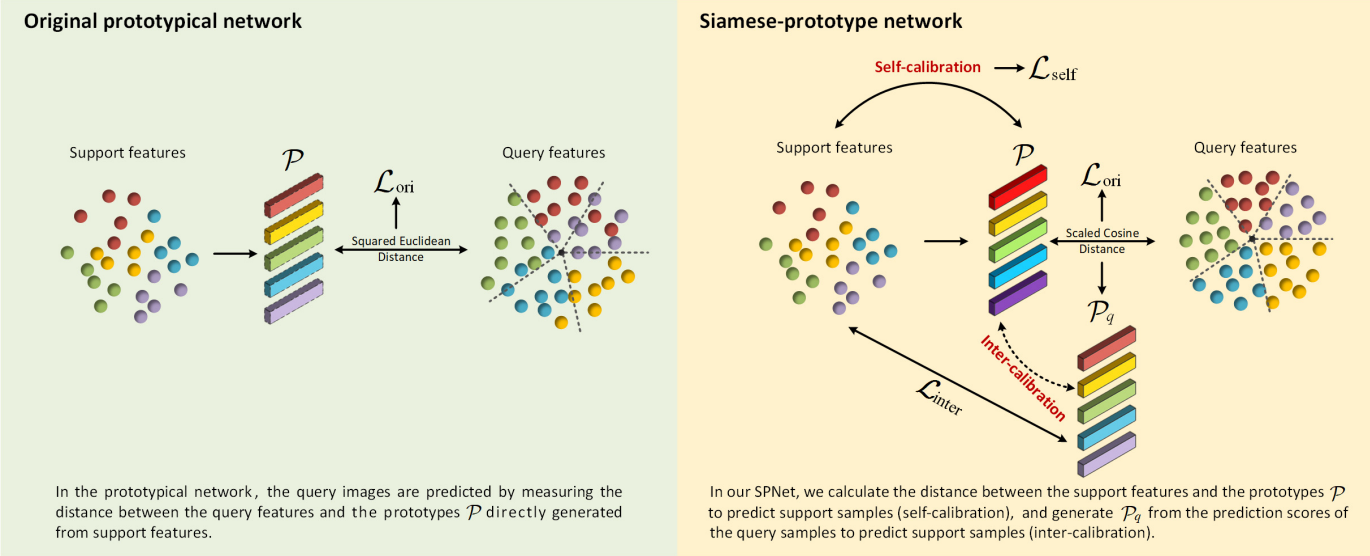


Fig. 1. Main innovations compared with the prototypical network. In the prototypical network, the query images are predicted by measuring the distance between the query features and the prototypes $\mathcal{P}$ directly generated from support features. In our SPNet, we calculate the distance between the support features and the prototypes $\mathcal{P}$ to predict support samples (SC) and then generate $\mathcal{P}_q$ from the prediction scores of the query samples to further calibrate the prototypes $\mathcal{P}$ (IC).

discover discriminative regions in the remote sensing images automatically. Zhang *et al.* [37] utilized metric learning to conduct the few-shot remote sensing image scene classification in which cosine distance was used with a learnable scale parameter.

The development of the few-shot remote sensing image scene classification is now in its initial rising stage but still quite promising. Among the classical few-shot classification methods, the prototypical network proposed by Snell *et al.* [25] is simple in principle but effective in practice, which has derived a series of studies based on it [34], [39]–[41]. These research works use either primitive knowledge or attention mechanism to refine the prototypes, bringing a mass of extra computation or memory usage, which we think is less than efficient. What is more, they directly generate prototypes and then figure out how to make it up, neglecting to figure out how to generate accurate prototypes from the root. The methodology of the prototypical network also inspires our work. However, on the contrary, we manage to generate accurate prototypes from the root without increasing the network architecture or data complexity. We learn useful information from what already exists in the baseline model without adding any learnable parameters. The information is embedded into the global loss function so that the entire model can be trained end-to-end, with no extra network layer or module added.

To be specific, we propose a Siamese-prototype network (SPNet) with prototype self-calibration (SC) and inter-calibration (IC) based on the prototypical network. Compared with the prototypical network, our main innovations are shown in Fig. 1. As mentioned before, we try to find a way to get more representative prototypes without increasing model computation and data footprint. An intuitive way is to add a constraint to the loss function to calibrate the prototypes during the training phase. Similarly, we utilize a certain distance metric between the prototypes and the support features to classify support samples. The cross-entropy loss is served as the constraint. Consequently, the unsupervised learning of prototypes evolves into supervised learning with no parameters to suffer. We call this process SC because the support features help to calibrate the prototypes generated from support features. Meanwhile, with the calibrated precise prototypes as a basis, we found that the query samples' prediction probabilities are valuable. Inspired by the thought of category mean [25], [37], we regard the query samples' prediction probabilities as the category means that carry the information for both the support and query sets. We denote them as $\mathcal{P}_q$ for convenience. $\mathcal{P}_q$ are regarded as another type of prototypes, which are also used to predict support samples in the training phase. As a result, the cross-entropy loss of this prediction serves as the second auxiliary constraint. We found in the experiments that $\mathcal{P}_q$ are effective for further calibration of original prototypes $\mathcal{P}$. The third loss promotes the mutual calibration of the two prototypes, so we call this IC process. The proposed model is trained end-to-end with these three losses, and the experimental results show that our model is simple while effective.

All in all, our contributions are as follows.

1) Based on the prototypical network, we propose a novel SC method for prototypes, which can bring more supervision to learn accurate prototypes without any learnable parameters added.
2) Through an IC method, we can obtain more accurate prediction probabilities for query samples and thus implicitly further calibrate the prototypes. The IC of the two Siamese prototypes provides more representative information for the model to make a better classification.
3) We demonstrate our method on three public remote sensing datasets, and the results show outstanding

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performances of our model in the few-shot remote sensing image scene classification task.

The remainder of this article is organized as follows. In Section II, we discuss some works related to few-shot learning and remote sensing image scene classification. Our method is introduced in Section III, and the experiments are described in Section IV. Finally, in Section V, we draw our conclusion.

II. Related Work

We will review some related works about few-shot learning, metric learning, and remote sensing image scene classification in this section.

A. Few-Shot Learning

Few-shot learning aims to recognize unlabeled samples from unseen classes given only a few labeled samples. There is an inseparable relationship between few-shot learning and meta-learning because most of the few-shot learning methods adopt a meta-learning strategy [19], [21]–[37], [39]–[42].

It is worth noting that few-shot learning has been introduced to remote sensing images more recently. For example, Gao *et al.* [43] proposed implementing the hyperspectral image classification in the case of few-shot learning. Zhai *et al.* [29] proposed an optimization-based few-shot classification method on remote sensing images. Li *et al.* [33] proposed a method similar to relation network for remote sensing image scene classification. Following closely behind, Li *et al.* [36] adopted the attention technique to discover discriminative regions in the remote sensing images automatically. Zhang *et al.* [37] utilized metric learning to conduct the task of few-shot remote sensing image scene classification in which cosine distance was used with a learnable scale parameter. Lately, zero-shot classification was exploited in remote sensing images [44]. The researchers collected a new remote sensing scene dataset including class-level semantic representations and alleviated the problem of class structure inconsistency between two hybrid spaces. In the future, zero-shot remote sensing image processing may be a new research hotspot.

B. Metric Learning

There are tens of thousands of parameters in the neural network classifier to be optimized. Therefore, if the ordinary neural network classifier is trained in the task of few-shot learning, it will be almost certainly overfitting. However, many nonparametric metric-based methods (e.g., nearest neighbor, K-nearest neighbor, and K-means) do not need to optimize parameters so that an end-to-end few-shot classifier can be constructed under the framework of meta-learning. The principle of this method is to model the distance distribution between samples, making similar samples close together and different samples far away.

Metric-based few-shot learning is early studied by Koch *et al.* [35] via Siamese networks. Soon afterward, Vinyals *et al.* [24] proposed the matching network that builds different encoders for the support set and the query set. The classifier's output is a weighted sum of the predicted values between the support samples and the query samples. In 2017, Snell *et al.* [25] proposed to calculate the mean of support samples for each category in the embedding space, which is seen as the prototype of the category. Then, the prediction was made by calculating the distance between the test sample and each prototype. Afterward, Sung *et al.* [26] argued that a proper metric is essential in the network, and it needs to be modeled. Their proposed network does not satisfy a single and fixed distance measurement but trains a network (e.g., CNN) to learn the distance measurement.

C. Remote Sensing Image Scene Classification

As an important part of remote sensing image understanding, remote sensing image scene classification has a steady development [45]–[54].

More recently, there has been much impressive work. Chaib *et al.* [14] enhanced scene classification accuracy by fusing features extracted from CNNs. He *et al.* [15] proposed a novel multilayer stacked covariance pooling (MSCP) algorithm for remote sensing image scene classification, which can combine multilayer feature maps extracted from pretrained CNN automatically. Cheng *et al.* [9] designed a novel objective function for learning discriminative CNNs (D-CNNs). The D-CNNs show better discriminability in scene classification. Liu *et al.* [11] introduced a hierarchical Wasseratein loss function to improve CNNs' discriminatory ability. Minetto *et al.* [16] devised a new remote sensing image scene classification framework named Hydra, an ensemble of CNNs which achieves excellent results on the NWPU-RESISC45 [55] dataset. Liu *et al.* [17] proposed the multiscale CNN (MCNN) to tackle the object scale variation problem in remote sensing scenes. One can refer to the recent survey paper [56] for more information about remote sensing image scene classification.

III. Method

A. Preliminary Knowledge

1) Prototypical Networks: As a metric-based few-shot learning method, it aims to calculate the mean of support samples for each category in the embedding space, which is seen as the prototype of that category, and the prediction was made by calculating the distance between the test sample and each prototype [25].

2) Meta-Learning: Existing meta-learning methods adopt episodic training manner. Generally speaking, the training set is randomly sampled into tens of thousands of meta tasks. Each task contains C categories with K samples for each category, forming a support set with $C * K$ samples. A batch sample is then extracted from the remaining data in these C classes as the task's query set. In each episode, it will sample different meta tasks. The model is required to learn how to distinguish the C categories from the $C * K$ data in each episode. Such a task is called the C-way K-shot problem. As episodes increase, the model learns the common parts in different meta tasks, such as extracting important features, comparing the similarity of samples, and forgetting the task-related parts

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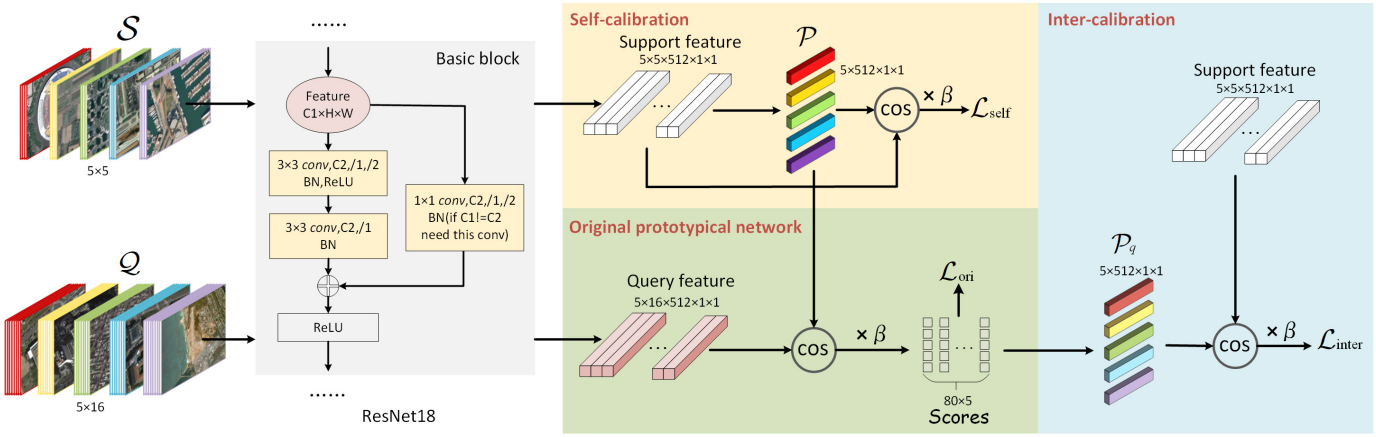


Fig. 2. Overall architecture of our SPNet in the case of five-way five-shot. In SC, we calculate the distance between the support features and the prototypes $\mathcal{P}$ to predict support samples. In IC, we generate $\mathcal{P}_q$ from the prediction scores of the query samples to predict support samples.

in meta tasks. The models learned through this learning mechanism can also perform better in the face of new meta tasks that are not seen before.

B. Method Overview

As shown in Fig. 2, our method is mainly based on the prototypical network with a ResNet-18 as the feature extractor. During an episode in the original prototypical network (the part framed in a green rectangle), the support images and each query image are first sent to the same feature extractor. The query image is then predicted by measuring the distance between the query feature and the prototypes generated from support features.

Our innovations are framed in the pink and blue rectangles, which are called SC and IC, respectively. To promote the generation of high-capacity prototypes, we propose to make full use of the supervision information from support samples. Concretely, we calculate the distance between the support features and the prototypes to predict support samples, thus utilizing another SC loss ($\mathcal{L}_{\text{self}}$ in Fig. 2) to guide the model to calibrate the prototypes. This process is called SC because the information from the support samples calibrates the prototypes generated from support samples themselves. The calibrated prototypes are more representative compared to the ones without supervision in the original procedure. Subsequently, with calibrated precise prototypes as a basis, one intuition is that the query samples' prediction probabilities are worth digging into. Inspired by the thought of category mean [25], [37], we consider the query samples' prediction probabilities as another type of prototypes ($\mathcal{P}_q$ in Fig. 2) that carry the information for both the support and query samples. Then, $\mathcal{P}_q$ are used to guide the classification of support images. Thus, the third IC loss ($\mathcal{L}_{\text{inter}}$ in Fig. 2) can be obtained. The IC loss can promote the mutual calibration of the two prototypes, so we call this IC process. Finally, the network updates gradient through a linear combination of these three losses.

We perform meta-learning on the model, which consists of the meta-training stage and meta-testing stage. In our method, the remote sensing image dataset is split into three subsets: the training set D_{train}, the validation set D_{val}, and the testing set D_{test}. The training set D_{train} and the validation set D_{val} are used in the meta-training stage, and the testing set D_{test} is used to test the trained model in the meta-testing stage. Note that the category space of each subset is disjoint, and we argue that the comparisons with other methods are convincing only if the category spaces of the three subsets are totally consistent.

We train the model in the widely used episodic training manner. In a C-way K-shot problem, we test the model with only K labeled samples given for each class in the meta-testing stage. Therefore, in the meta-training stage, we randomly sample C classes with K labeled samples for each class in each episode to imitate the settings in the meta-testing stage. The goal is to train a model that minimizes the C-way prediction loss in each episode. D_{train}, D_{val}, and D_{test} all consist of a mass of episodes. In each episode, except for $C * K$ labeled samples making up a support set $\mathcal{S}$, dozens of the remaining samples from the C classes form a query set $\mathcal{Q}$. Namely, $D_{\text{train}} = \{(\mathcal{S}_i, \mathcal{Q}_i)\}_{i=1}^{N_{\text{train}}}$ and $D_{\text{test}} = \{(\mathcal{S}_i, \mathcal{Q}_i)\}_{i=1}^{N_{\text{test}}}$, where N_{train} and N_{test} denote the numbers of episodes for training and testing, respectively.

C. Original Prototypical Network

In a C-way K-shot setting, given an episode with the support set $\mathcal{S} = \{(x_i, y_i)\}_{i=1}^{C*K}$, where each x_i is the support RGB image and $y_i \in \{0, 1, \ldots, C-1\}$ is the corresponding label. We denote the subset of $\mathcal{S}$ with all samples in class c as $\mathcal{S}_c = \{(x_j, y_j)\}_{j=1}^{K}$. We can directly get the embedded support features through the feature extractor f_θ. Subsequently, we compute the class prototype for class c as the mean of embedded support features in $\mathcal{S}_c$ as follows:

$$\mathcal{P}_c = \frac{1}{K} \sum_{x_i \in \mathcal{S}_c} f_\theta(x_i). \tag{1}$$

Note that in the C-way one-shot setting, the support feature of each class is directly considered as the prototype of that class because only one sample is supported for each class in an episode.

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For the query set $\mathcal{Q} = \{(q_i, y_i^q)\}_{i=1}^{N_q}$ in the episode, where N_q is the number of query samples, q_i is query image, and $y_i^q \in \{0, 1, \ldots, C-1\}$ is the corresponding label, we calculate the distance between the query features and the prototypes followed by a softmax function to predict the query images. Formally, the probability of predicting a query sample $q_i \in \mathcal{Q}$ as class c can be written as

$$p_\theta^i(y = c|q_i) = \frac{\exp\big(-\beta \cdot d(f_\theta(q_i), \mathcal{P}_c)\big)}{\sum_{c'}^{C} \exp\big(-\beta \cdot d(f_\theta(q_i), \mathcal{P}_{c'})\big)} \tag{2}$$

where $d(\cdot)$ represents a certain distance measurement and β is a scale parameter. Note that Snell *et al.* [25] used a squared Euclidean distance metric in the prototypical network, but we selected the scaled cosine distance as the measurement. On the one hand, Oreshkin *et al.* [42] proved that the squared Euclidean metric is superior to the cosine metric in [25] due to the influence of the different scaling of the metrics on the softmax function, and multiplying the cosine metric by a scaling factor is equivalent to squared Euclidean measures. On the other hand, in the experiments, we found that the cosine metric performs more robustly than the Euclidean metric in most cases. From our experiments, we can achieve the best performance when setting β to 50.

Then, the original loss for each episode can be computed as

$$\mathcal{L}_{\text{ori}} = -\frac{1}{N_q} \sum_{i=1}^{N_q} \log\big(p_\theta^i(y = c|q_i)\big) \tag{3}$$

where c denotes the correct label of corresponding query sample q_i.

In the meta-testing stage, the original prototypical network framework is adopted to output the classification results for test samples.

D. Self-Calibration

In the original prototypical network explained above, the prototypes are simply obtained from support features and directly used to predict query samples. We think that the flow of information in this process is insufficient to generate effective prototypes. Therefore, we propose an SC mechanism to make further use of the information from support samples. Specifically, the support samples in support set $\mathcal{S}$ are predicted in the same way as predicting query samples. The prediction loss helps to guide the model to calibrate the prototypes. Analogously, the prediction probability can be formulated as follows:

$$p_\theta^i(y = c|x_i) = \frac{\exp\big(-\beta \cdot d(f_\theta(x_i), \mathcal{P}_c)\big)}{\sum_{c'}^{C} \exp\big(-\beta \cdot d(f_\theta(x_i), \mathcal{P}_{c'})\big)} \tag{4}$$

where $d(\cdot)$ represents the cosine distance, x_i is the RGB image in support set $\mathcal{S}$, and the scale parameter β is also set to 50. Intuitively, the loss of SC can be calculated as the average of the negative log probability of all support samples in the episode

$$\mathcal{L}_{\text{self}} = -\frac{1}{C * K} \sum_{i=1}^{C*K} \log\big(p_\theta^i(y = c|x_i)\big). \tag{5}$$

In the C-way one-shot setting, the support feature of each class is directly considered as the prototype of that class, so the prototype SC is inoperative under the case of one-shot setting.

E. Intercalibration

So far, the information from given samples has not been fully mined. With the calibrated precise prototypes as a basis, we try to take advantage of the information in the prediction probabilities of query samples. To this end, we came up with a novel idea to predict support samples based on the query samples' information. Concretely, we treat the predicted probability of each class of the query samples as another prototype ($\mathcal{P}_q$) for that class, and we use these prototypes as a guide to reversely predict support samples in the same way as predicting query samples in the original prototypical network. $\mathcal{P}_q$ generated in this way contain the information of not only support features but also query features, and the prediction based on $\mathcal{P}_q$ implicitly corrects the prototypes by influencing the model's prediction of a query. More precisely, with the guarantee of end-to-end training, the effects between the two prototypes are reciprocal, that is to say, the two prototypes interact with each other in this process. The experimental results show that this interaction is positive and can force the model to achieve better performance. Generally, we call this IC for prototypes, which can be formulated as

$$\tilde{p}_\theta^i(y = c|x_i) = \frac{\exp\big(-\beta \cdot d(f_\theta(x_i), \mathcal{P}_q^c)\big)}{\sum_{c'}^{C} \exp\big(-\beta \cdot d(f_\theta(x_i), \mathcal{P}_q^{c'})\big)} \tag{6}$$

where

$$\mathcal{P}_q^c = \frac{1}{N_q^c} \sum_{i=1}^{N_q^c} p_\theta^i\big(y = c|q_i^c\big) \tag{7}$$

where N_q^c denotes the number of query samples with class c in the query set $\mathcal{Q}$, q_i^c denotes a query sample from class c, and $\mathcal{P}_q^c$ denotes the prototype $\mathcal{P}_q$ of class c, which is calculated by averaging all the probabilities that the query sample is predicted to be of class c.

Thus, we have the IC loss for the meta-training phase

$$\mathcal{L}_{\text{inter}} = -\frac{1}{C * K} \sum_{i=1}^{C*K} \log\big(\tilde{p}_\theta^i(y = c|x_i)\big). \tag{8}$$

The overall loss function presents in the form of linear combination of the aforementioned three losses

$$\mathcal{L} = \mathcal{L}_{\text{ori}} + \lambda_1 \mathcal{L}_{\text{self}} + \lambda_2 \mathcal{L}_{\text{inter}} \tag{9}$$

where λ_1 and λ_2 are the hyperparameters adjusting the weights of the two auxiliary losses, and the settings of them will be explained later.

F. Network Architecture

Our model is simple yet powerful, and there are no other learnable parameters than the one inherent in the backbone.

We perform our method mainly on ResNet-18 [57] backbone, and no other mechanisms or layers added. In fact, several few-shot learning methods use a shallow network to avoid

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overfitting, but inspired by [58], using a deeper backbone can significantly improve performance on the datasets with small domain shifts (i.e., domain differences between novel classes and base classes are small). Remote sensing images are relatively similar, unlike natural images with a large domain difference. Therefore, we try to use deeper backbones to extract representative features. The experimental results show that ResNet-18 [57] achieves better performance than shallow backbones such as the ones with four or five layers.

G. Computational Analysis

In this work, we emphasize the lightweight of our model. Without any attention mechanism or network layer added, our model simply captures useful information from what already exists in the baseline model. All parameters that need to be updated and learned in the training phase exist only in backbones. In the meta-testing phase, the trained model is used directly for testing without any subsequent processing or fine-tuning. As a result, compared with other methods, the increase in computation and memory usage in our method is very limited or even negligible.

IV. Experiments

In this section, we describe in detail the experiments carried out to demonstrate the effectiveness and efficiency of our method.

A. Datasets Description

We carried out experiments on three of the most commonly used public datasets: NWPU-RESISC45 dataset [55], WHU-RS19 dataset [59], and UC Merced dataset [60].

The NWPU-RESISC45 dataset was first proposed by the researchers of Northwestern Polytechnical University (NWPU) and then widely used as a benchmark for remote sensing image scene classification. This dataset consists of 45 scene categories, each with 700 RGB images, making up 31 500 RGB images with the size of 256×256. In our experiments, we split the dataset into 25, 10, and 10 categories for training, validation, and testing, respectively. For the sake of fair comparison, we adopt the same split as in [36].

The WHU-RS19 dataset is a widely used benchmark for remote sensing image scene classification, published by Wuhan University (WHU) in 2012. The dataset contains 19 scene categories, each with 50 or more images, for a total of 1005 RGB images. The size of each image is 600×600 pixels. We split the WHU-RS19 dataset with nine classes for training, five classes for validation, and five classes for testing. The split is also consistent with [36].

The UC Merced dataset is a remote sensing image dataset with 21 scene classes, and there are 100 RGB images per class, forming a total of 2100 images with an image size of 256×256. The dataset is released by the UC Merced computer vision laboratory and is used for land use classification research far and wide. For this dataset, we set the same split as [36] as well, in which ten classes as the training set, six classes as the validation set, and five classes as the testing set.

B. Implementation Details

Following the most commonly used meta-learning protocol, we perform five-way one-shot and five-way five-shot classification. We implement our model in the PyTorch framework and run our experiment on an NVIDIA GeForce GTX GPU. We use an adaptive momentum estimation algorithm (ADAM) to optimize our model, and the initial learning rate is set as 3×10^{-4}.

We use ResNet-18 as the feature extractor in our model. For each dataset, we train the model from scratch without using pretrained weights. All the images are randomly cropped to 224×224, and then, randomly horizontal flipping, brightness enhancing, color enhancing, and contrast enhancing are performed as image augmentation. We use global average pooling as the last layer for all the feature extractors, that is to say, the output feature is 1×1 in the spatial dimension. β is set to 50 in all the three procedures. In the five-shot case, λ_1 is set to 3 and λ_2 is set to 7. In the one-shot case, λ_1 is set to 1 and λ_2 is set to 3. The settings of these hyperparameters are explored in the following ablation study. In the meta-testing phase, neither the IC nor the SC is used, and only the original procedure is performed to output the classification results for testing images. In addition, the testing samples from the novel classes are independent of each other, that is to say, we do inductive inference. We use the overall accuracy to measure the performances of different methods, which is the ratio of all correctly classified images to the total images. We report the mean accuracy over 600 randomly generated tasks and a 95% confidence interval for all our experiments.

C. Experimental Results and Comparisons

We compare the classification accuracies of our model with that of several representative few-shot learning methods, including matching networks [24], prototypical networks [25], relation network [26], MAML [27], and Meta-SGD [30]. Note that our method is based on [25], so we will use the prototypical network with the same feature extractor as our baseline model. LLSR [29] is also an outstanding work on few-shot remote sensing image scene classification. Besides, Li *et al.* [36] introduced the attention mechanism into the feature learning module to learn more discriminative feature representations and adopted an adaptive matcher to robustly measure the similarity scores between remote sensing images, which is one of the state-of-the-art methods in the field of few-shot remote sensing image scene classification at present. We compare our method with their DLA-MatchNet in the same dataset split settings to get convincing comparison results.

Except for LLSR [29] and the DLA-MatchNet [36], all experimental results are conducted or reimplemented by ourselves. The accuracies of LLSR [29] and DLA-MatchNet [36] are directly borrowed from their reports. In addition to using the backbone same as that in the respective original paper for each method, we also verify the effect of the same backbone as our method in other methods, thus drawing a more convincing conclusion. Thereinto, the methods with "†" indicate that the backbone is ResNet-18, which is consistent with ours,

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TABLE I

FIVE-WAY CLASSIFICATION ACCURACIES OF DIFFERENT METHODS ON THE NWPU-RESISC45 DATASET. THE METHODS WITH "†" INDICATE THAT THE BACKBONE IS ResNet18, WHICH IS CONSISTENT WITH OURS, WHILE THE ONES WITHOUT "†" INDICATE THAT THE BACKBONE IS CONSISTENT WITH THE ORIGINAL PAPERS

Category	Method	1-shot	5-shot
Optimization-based	MAML [27]	37.36%±0.69%	45.94%±0.68%
	Meta-SGD [30]	60.63%±0.90%	75.75%±0.65%
	LLSR [29]	51.43%	72.90%
Metric-based	MatchingNet [24]	54.46%±0.77%	67.87%±0.59%
	ProtoNet [25]	50.82%±0.84%	74.38%±0.59%
	RelationNet [26]	58.61%±0.83%	78.63%±0.52%
	MatchingNet† [24]	64.41%±0.86%	76.33%±0.65%
	ProtoNet† [25]	65.20%±0.84%	80.52%±0.55%
	RelationNet† [26]	60.04%±0.85%	80.39%±0.56%
	DLA-MatchNet [36]	**68.80%±0.70%**	81.63%±0.46%
	SPNet (ours)	67.84%±0.87%	**83.94%±0.50%**

TABLE II

FIVE-WAY CLASSIFICATION ACCURACIES OF DIFFERENT METHODS ON THE WHU-RS19 DATASET. THE METHODS WITH "†" INDICATE THAT THE BACKBONE IS ResNet18, WHICH IS CONSISTENT WITH OURS, WHILE THE ONES WITHOUT "†" INDICATE THAT THE BACKBONE IS CONSISTENT WITH THE ORIGINAL PAPERS

Category	Method	1-shot	5-shot
Optimization-based	MAML [27]	46.72%±0.55%	79.88%±0.41%
	Meta-SGD [30]	51.54%±2.31%	61.74%±2.02%
	LLSR [29]	57.10%	70.65%
Metric-based	MatchingNet [24]	60.60%±0.68%	82.99%±0.40%
	ProtoNet [25]	70.88%±0.65%	85.62%±0.33%
	RelationNet [26]	60.54%±0.71%	76.24%±0.34%
	MatchingNet† [24]	67.68%±0.67%	85.01%±0.38%
	ProtoNet† [25]	76.36%±0.67%	85.00%±0.36%
	RelationNet† [26]	65.01%±0.72%	79.75%±0.32%
	DLA-MatchNet [36]	68.27%±1.83%	79.89%±0.33%
	SPNet (ours)	**81.06%±0.60%**	**88.04%±0.28%**

while the ones without "†" indicate that the backbone is consistent with the original paper. SPNet denotes our proposed Siamese-prototype network.

The comparisons of the few-shot classification accuracies of different methods on the NWPU-RESISC45 dataset are shown in Table I. From Table I, we can come to the following conclusions. First, our SPNet has raised the baseline (i.e., ProtoNet†) by 2.64% in the case of five-way one-shot and 3.42% in the case of five-way five-shot. Second, we achieve the best performance for five-shot case and the second-best performance for one-shot case. Actually, we have surpassed all the optimization-based methods. We attribute the gap between our method and DLA-MatchNet in the one-shot case to our model's particularity. On the one hand, our model is essentially based on the prototypical network, whereas the DLA-MatchNet is based on the matching network. In the case of very few samples available, such as one-shot case, the prototype of the category degenerates to the global average pooling of a single sample, which brings a great challenge to the subsequent inferences based on prototypes. On the other hand, the class prototype is identical to the support feature of that class in the one-shot case, so their distance is 0. In other words, the SC does not work under such circumstance. Third, with the same network architecture (ResNet-18), our method goes far beyond the three classical few-shot classification methods (MachingNet†, ProtoNet†, and RelationNet†). Finally, the comparison of the first six rows of the table indicates that a deeper backbone may show better adaptability in few-shot scenarios of remote sensing images.

Table II shows the comparison of different methods on the WHU-RS19 dataset. Surprisingly, we have achieved the best classification performance and gone far beyond other methods, whether optimization-based or metric-based. Compared to the DLA-MatchNet [36] which is the second-best and also metric-based, our SPNet is almost 13% and 8% more accurate than it in the one-shot case and five-shot case, respectively. Besides, our SPNet performs much better than LLSR [29] in both the one- and five-shot cases, where LLSR [29] needs fine-tuning during the testing stage and our SPNet does not. Compared with the baseline model (ProtoNet†), we have surpassed it by about 5% in the one-shot case and 3% in the five-shot case.

TABLE III

FIVE-WAY CLASSIFICATION ACCURACIES OF DIFFERENT METHODS ON THE UCM DATASET. THE METHODS WITH "†" INDICATE THAT THE BACKBONE IS ResNet18, WHICH IS CONSISTENT WITH OURS, WHILE THE ONES WITHOUT "†" INDICATE THAT THE BACKBONE IS CONSISTENT WITH THE ORIGINAL PAPERS

Category	Method	1-shot	5-shot
Optimization-based	MAML [27]	43.65%±0.68%	58.43%±0.64%
	Meta-SGD [30]	50.52%±2.61%	60.82%±2.00%
	LLSR [29]	39.47%	57.40%
Metric-based	MatchingNet [24]	46.16%±0.71%	66.73%±0.56%
	ProtoNet [25]	52.62%±0.70%	65.93%±0.57%
	RelationNet [26]	48.89%±0.73%	64.10%±0.54%
	MatchingNet† [24]	48.18%±0.75%	67.39%±0.50%
	ProtoNet† [25]	53.85%±0.78%	71.23%±0.48%
	RelationNet† [26]	50.07%±0.72%	65.22%±0.52%
	DLA-MatchNet [36]	53.76%±0.62%	63.01%±0.51%
	SPNet (ours)	**57.64%±0.73%**	**73.52%±0.51%**

We also perform comparative experiments on the UCM dataset. As shown in Table III, our SPNet achieves the best performance among all the methods used for comparison. Being consistent with the other two datasets, the few-shot classification accuracies of our SPNet on the UCM dataset also outperform all the comparative methods with the same backbones. We have surpassed the baseline model (ProtoNet†) by 3.79% in the one-shot case and 2.29% in the five-shot case. Note that our method is superior to DLA-MatchNet [36] by 10.51% in the case of five-shot case and superior to

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LLSR [29] by 16%–18% without fine-tuning in the testing stage, which are big steps in the few-shot remote sensing image scene classification.

It is obvious that our SPNet performs much better on the UCM dataset and the WHU-RS19 dataset than on the NWPU-RESISC45 dataset. We attribute this phenomenon to the differences in datasets. The NWPU dataset is ten times bigger than the other two datasets, and compared with the other two datasets, it has more similar categories such as church versus palace and railway station versus stadium, which brings more challenges to discriminative Siamese-prototype generation.

In summary, on the one hand, our proposed SPNet with prototype SC and IC outperforms the baseline model to a great extent in all the cases while keeping the number of parameters and the FLOPs constant. This confirms that the SC and IC proposed in our model can effectively facilitate obtaining representative prototypes. On the other hand, our method is superior to other advanced methods to a great extent in most cases, including MatchingNet† [24], ProtoNet† [25], RelationNet† [26], MAML [27], Meta-SGD [30], LLSR [29], and DLA-MatchNet [36], where we use the same backbone as ours to reimplement MatchingNet†, ProtoNet†, and RelationNet† for fair comparisons. Moreover, there is no fine-tuning needed in the testing stage in our method. All these have demonstrated the simplicity of the scaled cosine distance metric-based learning and the effectiveness of SC and IC in our model.

In order to intuitively compare the performance of different methods, we randomly selected several images from some difficult categories to observe the classification results of different methods, as shown in the first four lines of Fig. 5. Our SPNet stands out compared to MAML, Meta-SGD, and ProtoNet†. However, when we encounter extremely difficult samples, our method also goes wrong, as shown in the last line of Fig. 3. The scenes of commercial areas are filled with a large number of dense buildings, interspersed with some trees and roads, just like the scenes of dense residentials. Similarly, the images of the runway and the freeway are both composed of a white road and a solid color background around it.

D. Ablation Study

In this section, we evaluated the effectiveness of our proposed prototype SC and prototype IC, and the influence of different distance metric and hyperparameters settings.

As shown in Table IV, baseline denotes the ProtoNet method with cosine distance as a metric and ResNet-18 backbone as the feature extractor, SC denotes SC, and IC denotes IC. We use ResNet-18 backbone in each situation, keeping $\beta = 50$ everywhere the cosine distance is used. In the case of adopting both SC and IC, we set $\lambda_1 = 3$ and $\lambda_2 = 7$ in the five-shot case and $\lambda_1 = 1$ and $\lambda_2 = 3$ in the one-shot case because the following experiments demonstrate that such a setting is the best. From Table IV, when SC is adopted alone, the baseline is improved evidently, with a 1.3% improvement in the one-shot case and 2.28% improvement in the five-shot case. When IC is used alone, the improvement is relatively slight, which is 0.58% in the one-shot case and 1.39% in the five-shot case. As expected, the model works best when SC and IC are adopted together, where 3.42% and 2.64% are improved against the baseline in the five- and one-shot cases, respectively.

The impact of different distance measures is shown in Table V. We report the classification accuracies of using squared Euclidean distance and scaled cosine distance on the three datasets. The settings of β, λ_1, and λ_2 and the feature extractor are the same as before. From Table V, we can see that when using squared Euclidean distance, the accuracies will always fall behind the scaled cosine distance. What is

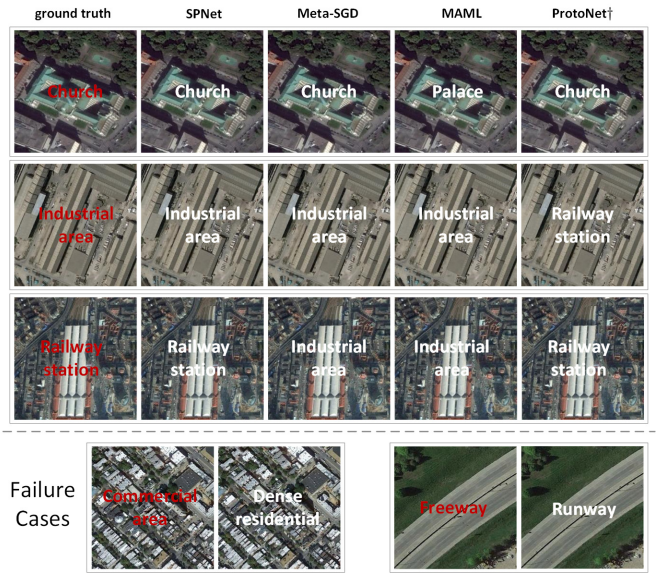


Fig. 3. Classification of hard samples by different methods and failure cases of SPNet.

TABLE IV
EFFECTS OF PROTOTYPE SC AND IC ON THE NWPU-RESISC45 DATASET

Method	1-shot	5-shot
Baseline	65.20%±0.84%	80.52%±0.55%
Baseline + SC	66.50%±0.85%	82.80%±0.51%
Baseline + IC	65.78%±0.87%	81.89%±0.52%
Baseline + SC + IC (ours)	67.84%±0.87%	83.94%±0.50%

TABLE V
IMPACT OF DIFFERENT DISTANCE MEASUREMENTS. EUCLIDEAN DENOTES THE SQUARED EUCLIDEAN DISTANCE AND COSINE DENOTES THE SCALED COSINE DISTANCE

Data sets	Metric	1-shot	5-shot
NWPU-RESISC45	Euclidean	65.68%±0.88%	81.69%±0.54%
	Cosine (ours)	67.84%±0.87%	83.94%±0.50%
WHU-RS19	Euclidean	76.08%±0.63%	84.86%±0.33%
	Cosine (ours)	81.06%±0.60%	88.04%±0.28%
UCM	Euclidean	55.79%±0.74%	69.52%±0.55%
	Cosine (ours)	57.64%±0.73%	73.52%±0.51%

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TABLE VI
IMPACT OF HYPERPARAMETER β ON THE NWPU-RESISC45 DATASET

β	1-shot	5-shot
10	65.14%±0.94%	77.91%±0.61%
20	66.49%±0.87%	81.40%±0.51%
50	67.84%±0.87%	83.94%±0.50%
100	67.82%±0.86%	82.69%±0.51%

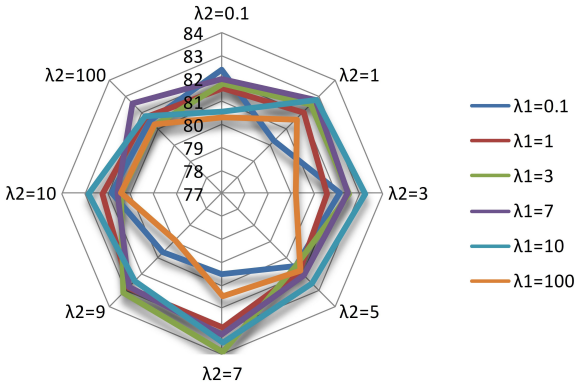


Fig. 4. Five-way five-shot classification accuracies on the NWPU-RESISC45 dataset under different settings of λ_1 and λ_2. The model performs best when $\lambda_1 = 3$ and $\lambda_2 = 7$.

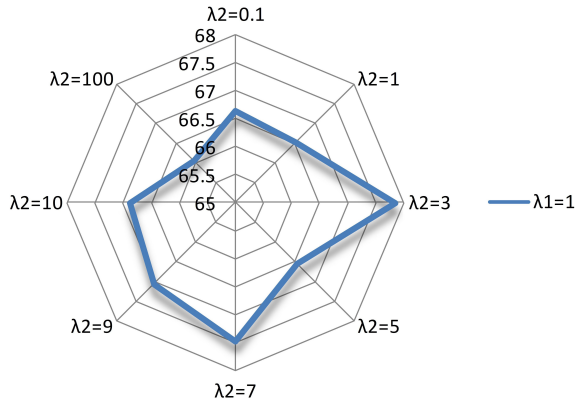


Fig. 5. Five-way one-shot classification accuracies on the NWPU-RESISC45 dataset under different settings of λ_2. The model performs best when $\lambda_2 = 3$.

more, the squared Euclidean distance behaves more unstable with a bigger variance, and sometimes, the accuracies even fall behind the baseline.

The influence of hyperparameter setting on few-shot classification performance is also studied. Thereinto, the impact of the scaled parameter β on classification is reported in Table VI. We use the same settings as in the "Baseline+SC+IC" in Table IV except for β. Then, we report the classification accuracies on the NWPU-RESISC45 dataset under different settings of β. Referring to the common settings in some literature works [19], [37], we set β to 10, 20, and 50 and found that 50 performs better. We then further tried 100 and did not get better results, so 50 is finally chosen. The results also show that it will get a big variance when β is too small.

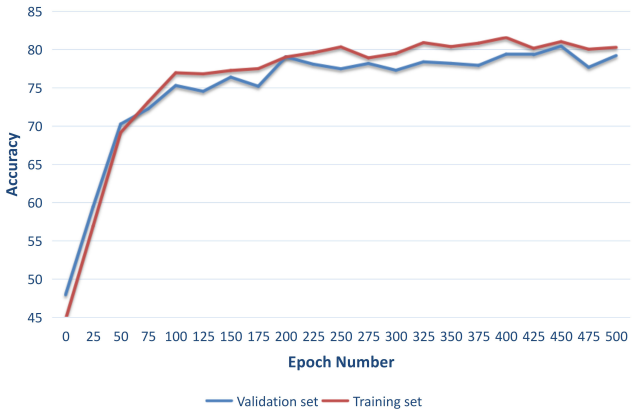


Fig. 6. Training and validation accuracy under the five-way five-shot setting on the NWPU-RESISC45 dataset.

Finally, we explored the impact of hyperparameters λ_1 and λ_2. We first explore it on a large scale by setting both of them as 0.1, 1, 10, and 100. As shown in Fig. 4, the model performs better when both parameters are in the range of 1–10. Therefore, we then tried many combinations of λ_1 and λ_2 in the interval [1, 10]. For visual clarity, Fig. 4 shows only part of the situation. In the five-way one-shot case, we fix λ_1 as 1 because the SC is inoperative in the one-shot case. Similarly, we set λ_2 in a wide range and analyze the impact. The results of five-way one-shot case are shown in Fig. 5.

Fig. 6 shows the accuracy of our SPNet on the training set and the validation set. The training process contains 500 epochs and each epoch contains 100 episodes. The highest validation accuracy appears in the 450th epoch.

V. CONCLUSION

This article proposed a simple yet effective method for the task of few-shot remote sensing image scene classification, called SPNet. First, we add a constraint into the loss function to utilize the supervision information from support samples to calibrate the prototypes. Second, with the calibrated accurate prototypes as a basis, we take full advantage of the representative information from prediction probabilities of query samples. We consider them as another type of prototypes that carry the information for both the support and query samples. According to our extensive experiments, both the SC and the IC are conducive to improving the accuracies of subsequent inference based on prototypes and achieving competitive performances. Simultaneously, no extra network layer, mechanism, or module is added, and our model is trained end-to-end without fine-tuning in the meta-testing stage.

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